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Numerical analysis of quenching distance in laminar premixed hydrogen and methane flames

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ABSTRACT

The quenching behavior of laminar premixed hydrogen-air and methane-air flames is studied in two-dimensional configurations with detailed chemistry. Quenching distances are determined by simulating an initially stationary flame and then decreasing the inlet speed, allowing upstream flame propagation in a converging duct. The quenching distance is then defined as the distance between the cold surfaces where the flame extinguishes due to heat loss to the walls. The results for methane-air and hydrogen-air mixtures at various equivalence ratios are compared with experimental data, showing good agreement. The effects of flow inlet velocity, geometry, and wall temperature on quenching distance are investigated. The quenching distance is found to be sensitive to the inlet speed and decreases with decreasing inlet speed. The quenching distance is shown to decrease linearly with increasing wall temperature in hydrogen-air flames. In addition, the quenching distance depends on the geometry of the setup. Slit and annular ducts result in similar quenching distances, whereas circular ducts have higher quenching distances compared to the others. The study highlights the importance of considering these factors in burner design and flashback prevention devices.

1. Introduction

With the aim of decarbonization, studies on alternative fuels to replace natural gas have accelerated in recent years, and hydrogen is one of the possible candidates [1,2,3]. When H_2 is generated using renewable energy, it becomes a promising substitute fuel to reduce carbon emissions in both domestic and industrial applications [4]. However, the fundamental characteristics of H_2 flames, such as laminar flame speed, adiabatic flame temperature, ignition energy, flashback/blow-off limits, and quenching distance, are substantially different from those of natural gas flames [5]. Due to their high reactivity and low ignition energy, H_2 flames tend to be perceived as more dangerous compared to conventional gaseous fuels like natural gas [6,7]. During the transition to a new fuel, extra care should be taken to prevent any disasters, making the process more complex than it otherwise would be. Therefore, a detailed investigation of the fundamental characteristics of H_2 flames is necessary. In recent years, studies on substituting natural gas with hydrogen, either fully or partially in premixed burners, have escalated with a focus on flashback, blow-off, and autoignition behavior [8,9,10,11,12]. In this work, the focus will be on the quenching phenomenon as well as the numerical determination of quenching distances of CH_4-air and H_2-air flames and their differences.

Flame quenching is a critical phenomenon in combustion, involving the extinction of flames when they come into contact with a cooler surface or enter a narrow passage like for instance a burner port. Quenching occurs when the local heat losses to the surrounding, usually conjugate heat transfer to the surrounding solids, is larger than the heat production in the flame, leading to a lower flame temperature and decreasing reaction rate. Furthermore, flames can be quenched when they experience a high aerodynamic stretch [13], but this effect is beyond the scope of the presented research. Understanding flame quenching is vital for several reasons, particularly safety and efficiency in the design of burners. In numerous applications, including gas turbines, internal combustion engines, and household gas appliances, preventing unintentional flame propagation is crucial to avoid accidents. It plays a key role in safety devices, such as flame arrestors and quenching meshes, which are designed to control and extinguish unintended flames to prevent explosions or damage to combustor parts. In the literature different types of quenching are mentioned, such as tube/slit quenching, head-on quenching, and sidewall quenching [14,15].

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This study focuses on tube/slit quenching configurations, which are more relevant for burner design applications. In this configuration, total flame quenching occurs in an enclosed area such as tubes, slits, annular passages, or parallel plates where the distance is sufficiently small [14]. In the literature, various methods have been reported to determine the quenching distance. The first method to be mentioned is the parallel plate/flanged electrode spark (FESI) method where the fuel–air mixture is ignited between two parallel plates with an ignition source. The quenching distance is the plate separation distance below which the mixture cannot be successfully ignited. This method is used by Lewis and von Elbe [5] for various mixtures including CH_4-air and H_2-air. Later, Hong et al. [16] investigated the quenching distances of H_2-air mixtures with steam addition using this method and reported that the quenching distance decreases with increasing initial pressure using FESI. Harris et al. [17] investigated the burning velocities and quenching distances of methane and propane mixtures with air and oxygen-enriched air. They obtained quenching distances in both parallel plates and circular tube configurations, and reported that quenching distance decreases with increasing oxygen content in the oxidizer. Jarosinski [18] investigated the quenching distance of CH_4-air mixtures in a square channel by inserting a block inside the channel to create a narrow gap. In the experiment, the stagnant premixture is ignited from the open end, and the flame propagates towards the block and finally quenches inside the narrow gap due to heat loss to the cold walls. Liu et al. [19,20] and Jung et al. [21] investigated laminar flame speeds and quenching distances of CH_4-air and H_2-air premixtures respectively with a method called annular stepwise diverging tube (ASDT). In this method, first the flame stabilizes between the outer tube and a stepwise diverging solid metal core by igniting the premixture which is supplied to the system with a constant flow rate. Then, the flow rate is reduced to a certain value, and the flame starts to travel upstream while the distance between the inner core and the outer tube decreases gradually. The distance between the outer tube and the inner core where the flame extinguishes is then marked as the quenching distance. In a recent work by Pers et al. [22], H_2 flame quenching onset of flashback in a perforated burner is studied, and the effect of the burner temperature on flame quenching is highlighted.

To the best of the author's knowledge, no numerical study has been reported on the quenching distances of H_2-air or CH_4-air mixtures using detailed chemistry for the configurations studied. With the help of a computational fluid dynamics approach, parametric studies can be performed efficiently without building expensive experimental setups. In addition, this approach can eliminate imperfections encountered in experimental setups, such as mixing issues, deviations in equivalence ratios, and flow perturbations due to manufacturing imperfections. In this study, the capability of accurately capturing the quenching distances is evaluated using a detailed chemistry approach, and the obtained results are compared with the experimental results found in the literature. A comprehensive understanding of how sensitive the quenching distance measurements are to setup conditions is still lacking in the literature. This study aims to fill this gap by highlighting the effects of different factors such as specific geometry, fuel velocity, and wall temperature on the quenching distance and the sensitivity of the obtained results. It should be noted that both "quenching distance" and "quenching diameter" are terms used in the literature for different configurations. For this work, the authors have chosen to use the term "quenching distance" for all configurations to prevent any confusion.

Details of the governing equations, modeling, and boundary conditions are introduced in the section on Numerical Modeling. Next, the process of determining quenching distances under various conditions is explained in the Methodology section. The effects of final velocity, geometry, temperature, and diffusion modeling are discussed in the Results and Discussion section. The quenching distance results are presented in two separate subsections for methane-air and hydrogen-air. In the Conclusions we summarize and close with some final remarks.

2. Numerical modeling

In this study, the reactive laminar flow is modeled using ANSYS Fluent 2023R1 [23]. The governing equations are solved in a finite-volume framework and the transport equations for the transient laminar flow and species are given below.

Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0 \tag{1}$$

Momentum Equation

$$\frac{\partial \rho \boldsymbol{v}}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}\boldsymbol{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} \tag{2}$$

Energy Equation

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (\boldsymbol{v}(\rho E + p)) = \nabla \cdot \left(k \nabla T + \sum_j h_i \boldsymbol{J_i}\right) - \sum_j h_i \dot{\omega}_i \tag{3}$$

Species Transport Equations

$$\frac{\partial (\rho Y_i)}{\partial t} + \nabla \cdot (\rho \boldsymbol{v} Y_i) = -\nabla \cdot \boldsymbol{J_i} + \dot{\omega}_i \tag{4}$$

Diffusion Flux

$$\boldsymbol{J_i} = -\sum_{j=1}^{N-1} \rho D_{ij} \nabla Y_j - D_{T,i} \frac{\nabla T}{T} \tag{5}$$

In this set of equations, $\mathbf{v}$ is the velocity field, ρ is the density, p is the pressure and τ is the stress tensor. E is the total energy per unit mass, k is conductivity, h_i is the specific enthalpy, Y_i is the mass fraction, and $\dot{\omega}_i$ is the net production rate of species i. The diffusion flux ($\mathbf{J}_i$) is modeled as the combination of multicomponent and Soret diffusion where D_{ij} is the generalized diffusion coefficient of species i into j. The thermal diffusion coefficient $D_{T,i}$ is modeled using the empirical formula of Kuo [24].

The pressure-based PISO algorithm is used with a second-order scheme for the solution of the system of transport equations. The chemical reactions are modeled using finite-rate chemistry with the stiff chemistry approach. For CH_4–air premixtures, both GRI 3.0 (53 species and 325 reactions) [25] and reduced DRM 19 (21 species and 84 reactions) [26] mechanisms are used, but no significant differences are observed in the quenching distance. Therefore, only DRM 19 results are reported in the manuscript. For H_2–air premixtures, both Burke [27] (9 species and 19 reactions) and Stanford (9 species and 20 reactions) [28] mechanisms are used. No significant differences are observed, especially for equivalence ratios below 1.6, and therefore only the Burke mechanism results are reported. The effect of the reaction mechanism on the quenching distance for the slit configuration is reported in Appendix A for both fuels.

Multicomponent diffusion is modeled using generalized diffusion coefficients, which are calculated by solving the Maxwell-Stefan equations [29], assuming that the ideal gas law is valid. Binary diffusion coefficients are calculated with the Chapman-Enskog [30] formula using Lennard-Jones parameters. Thermal diffusion effects are included since it is especially important for lean H_2-air mixtures where the Lewis number is far from unity [31,32]. Radiation effects are not taken into account in this study. The effects of conjugate heat transfer are neglected since the flame residence time in the channel is not long enough to heat up the burner walls during the quenching process. One-dimensional free flame simulations are performed using CANTERA [33] to obtain thermal flame thicknesses using the expression [14]

$$\delta_F = \frac{T_b^0 - T_u}{\left|\frac{dT}{dx}\right|_{\max}} \tag{6}$$

2.1. Computational domain and boundary conditions

Laminar flame quenching was studied in 3 different geometries to investigate the effect on the quenching distance. The configurations are shown schematically in Fig. 1. They correspond to a two-dimensional slit (A), a circular duct (B) and an annular duct (C), respectively. The width/diameter is increasing in downstream direction of the flow (which is indicated with arrows) in each configuration. For all configurations, a 2D modeling approach is used to decrease computational cost. The modeled 2D plane is indicated with a red plane in Fig. 1.

Fig. 2 represents the domain and boundary condition for configurations (A) and (B) and Fig. 3 shows them for configuration (C). The inlet velocity is modeled as a Poiseuille flow with

$$U_{\mathrm{ini}}\left[1-\left(\frac{2y}{d}\right)^{2}\right] \tag{7}$$

where U_{ini} is the velocity in the x-direction (flow direction) on the symmetry axis.

The pressure outlet is set to 0 kPa relative to the reference pressure of 100 kPa. The inlet and outlet dimensions (d_{in} and d_{out}) are varied for different equivalence ratios and fuels while keeping the slope angle constant (9.6 degree for CH_4 and 4.8 degree for H_2). This variation allows for capturing different quenching distances without increasing the computational domain, thus reducing computational costs. The walls are modeled as non-slip with an isothermal temperature. They are considered inert, and the species flux is set to zero. For CH_4, all calculations are performed at a wall temperature of 300 K, while the effect of wall temperature is investigated for H_2 flames, ranging from 300 K to 800 K in Section 4.2.3. The fresh mixture temperature at the inlet is kept equal to the wall temperature. Only half of the physical domain is modeled, with the axis set to planar symmetry for configuration (A) and axisymmetry for configuration (B). For configuration (C), the annular passage is modeled with a distance w from the axis of symmetry. The value of w is set to 55 mm to correspond to the outer diameter of the ASDT setup to compare the quenching distance (d_q) values. The domain length (L) is varied from 5 mm to 15 mm. After a mesh independence study, a nearly uniform hexagonal mesh with a 40 μm resolution was chosen for the CH_4-air mixtures and a 10 μm mesh resolution for H_2-air mixtures. Transient simulations are performed with a 1 μs time step size to ensure sufficient resolution at the instance of quenching. Table 1 summarizes the test cases for different fuels, geometries, final velocities (U_f), wall temperatures (U_{wall}), and equivalence ratios (ϕ).

At first, a steady-state cold flow is modeled with an initial fully developed velocity profile with value U_{ini} at the symmetry axis. The mixture is then ignited by introducing a temperature patch higher than the ignition temperature close to the outlet, and the governing equations are solved until a steady flame solution is obtained. Since the laminar flame speed and reactivity changes with equivalence ratio, U_{ini} is varied for different configurations to obtain a stable flame on the downstream side of the quenching distance.

In the literature, various studies [35,36,37] report a peak in heat release rate near the walls during H_2-air premixed flame modeling. This behavior has been observed in studies using a finite rate chemistry approach. In the current study, this effect was found to be enhanced when multicomponent diffusion is modeled, particularly for rich mixtures. This peak is associated with the accumulation of H_2O_2 and HO_2 radicals near the wall [35,38]. Since these intermediate reactions do not have any activation energy, it was observed that H radicals convert into H_2O_2 and HO_2 in the cold region near the wall, leading to a peak in the heat release rate near the walls. In this study, this heat release rate peak was also observed because the walls were modeled as inert and a detailed finite-rate chemistry approach was used. To determine whether this phenomenon affects the results of the quenching distance, the mass fractions of the radicals involved in these reactions (H, OH, H_2O_2 and HO_2) were set to zero on the wall in the slit configuration. No significant differences in the quenching distance were found. Therefore, although this heat release rate peak might affect other aspects of flame dynamics, it does not impact the scope of this study.

3. Methodology

As explained in the previous chapter, first a steady-state solution is obtained with a stationary flame. This solution is then used as the initial condition for the transient simulations. The velocity U_{ini} is decreased to a value U_f (final velocity) which allows the flame front to travel in the upstream direction towards the inlet and quench at a certain distance. The main reason behind quenching is that the flame loses more heat toward the walls than it produces with chemical reactions. Then, the flame cools down, while it propagates upstream, and finally gets too cold to sustain the chemical reactions. The wall-to-wall distance of the channel at the point where the flame front is unable to sustain itself and quenches is marked as the quenching distance (d_q).

The flame front is represented by the net production rate ($\dot{\omega}_i$) of CH_4 or H_2 depending on the fuel. Because the reactivity of the mixture is lower near the cold walls compared to the symmetry axis, the flame speed decreases locally and a convex flame shape is observed rather than a flat flame. Fig. 5 shows the quenching process in which the flame loses its energy to the cold walls and the methane consumption rate decreases. During the quenching process, the flame front stops traveling and becomes rapidly weaker at that axial location. As the flame gets weaker, it moves slightly downstream because it cannot withstand the upcoming flow due to the decreasing reaction rate. Eventually, the flame then quenches and the remaining radicals are transported downstream with the flow.

To determine the quenching distance, the minimum $\dot{\omega}_{H_2}$ is tracked along the symmetry axis, and its location is considered as the position of the flame front. Negative values of $\dot{\omega}_{H_2}$ indicate that hydrogen is consumed, therefore the minimum $\dot{\omega}_{H_2}$ represents the strongest reacting

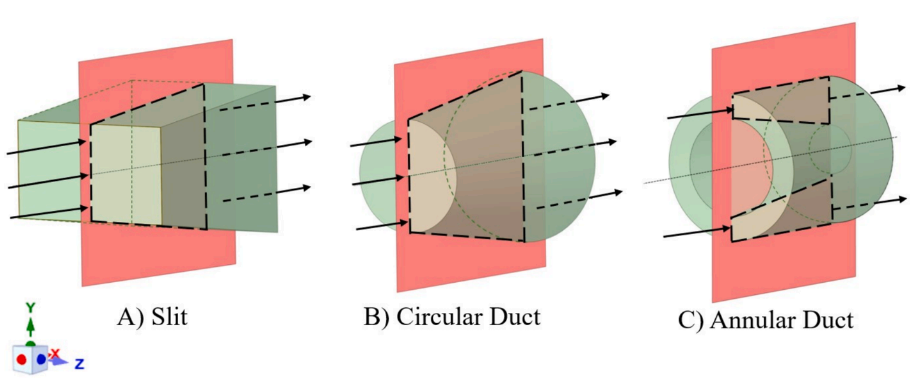

Fig. 1. Different geometries investigated in this study: slit, circular duct and annular duct.

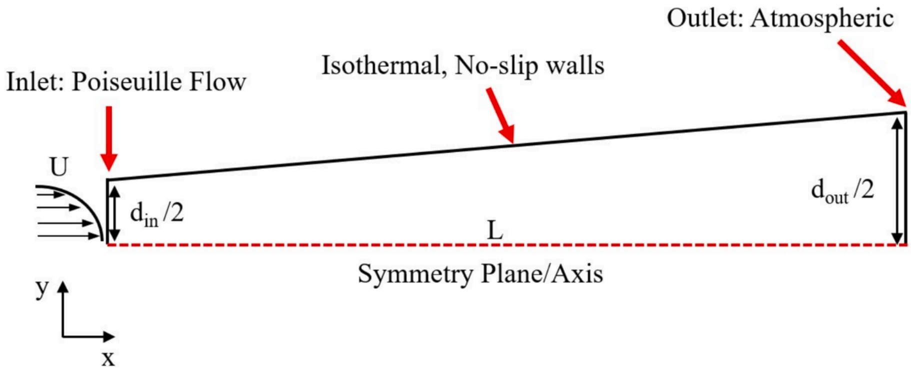

Fig. 2. Boundary conditions and computational domain for slit and circular duct geometries.

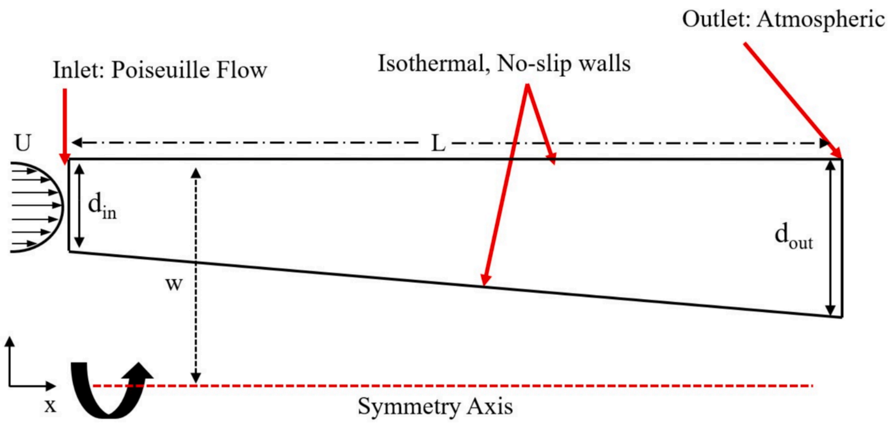

Fig. 3. Boundary conditions and computational domain for the annular duct geometry.

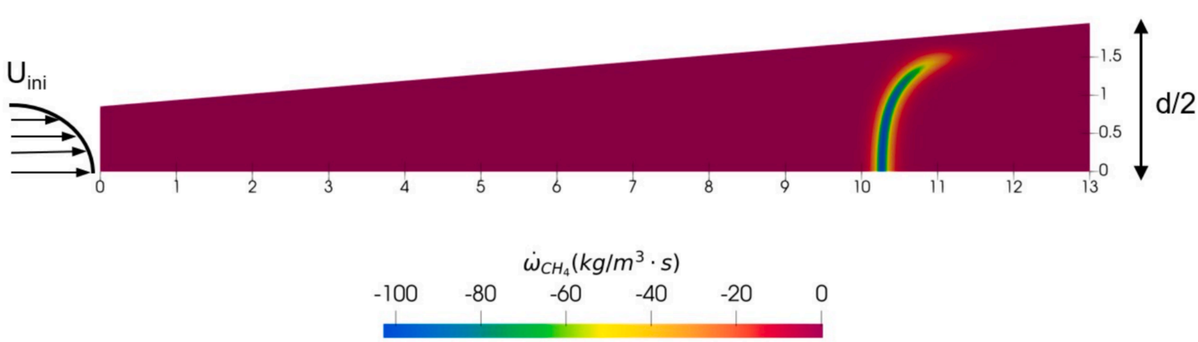

Fig. 4. Steady methane-air flame in the slit geometry for ϕ = 1 at T = 300 K.Fig. 4 shows a stable flame solution with a convex shape towards the upstream for the CH_4-air mixture at an equivalence ratio of ϕ = 1. This flame shape has also been observed in previous studies on flames in confined channels [34].

Table 1
Summary of test cases for different fuels, geometries, final velocities (U_f), wall temperatures (T_{wall}), and equivalence ratios (ϕ).

Fuel	Geometry	Final Velocity (U_f)	Wall Temperature (T_{wall})	Equivalence Ratio (ϕ)
H_2	Slit	0, 1, 3 m/s	300–800 K	0.5–2.4
	Circular Duct	0	300 K	0.5–2.4
	Annular Duct	0	300 K	0.5–2.4
CH_4	Slit	0, 0.25 m/s	300 K	0.7–1.3
	Circular Duct	0	300 K	0.7–1.3
	Annular Duct	0	300 K	0.7–1.3

part of the flame front. Fig. 6 shows the transient location of the flame front converted to the local distance between the walls and the corresponding source term $\dot{\omega}_{H_2}$ for an H_2-air flame under stoichiometric conditions with $T_{\mathrm{wall}} = 300$ K for the slit configuration. We identify three main regions in the quenching process: The initial phase, the phase of steady heat loss and the rapid quenching phase. The instance (a) in Fig. 6 marks the location of the stationary flame before the velocity decrease at the inlet. As the flame front starts to move upstream, the consumption of hydrogen increases initially between $d = 0.625$ mm and $d = 0.675$ mm. In instant (b), the consumption rate decreases monotonically as the heat loss to the wall increases. After instant (b), the consumption rate $\dot{\omega}_{H_2}$ decreases faster and finally rapid quenching begins after instant (c), where the consumption rate quickly reduces to zero while the location of the flame hardly changes. Around the quenching point, the flame front loses its energy without changing its location, although in the final

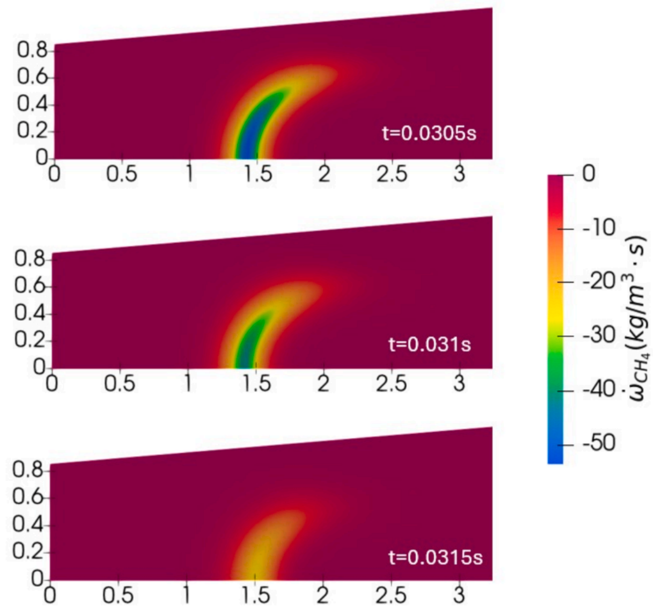

Fig. 5. Fuel consumption rate $\dot{\omega}_{CH_4}$ during the quenching process of a CH_4-air flame for $\phi = 1$ at T = 300 K.

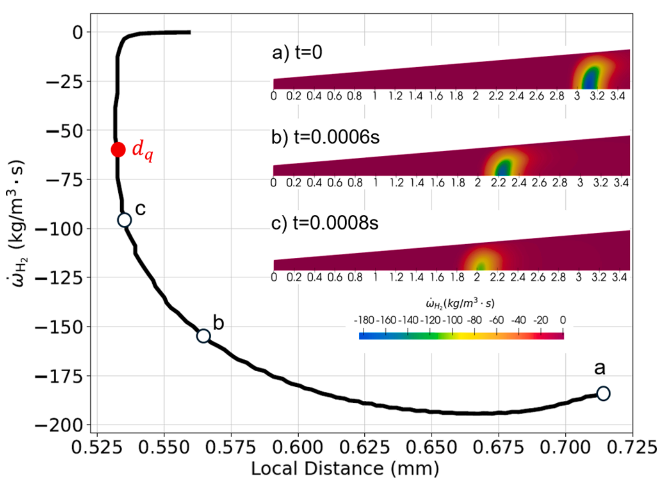

Fig. 6. Transient quenching behaviour of H_2-air flame in a slit for $\phi = 1$, Uini = 7 m/s, Uf = 0, at T = 300 K, showing minimum $\dot{\omega}_{H_2}$ local width. The insets show the source term distributions at the instances (a), (b), and (c).

quenching there is a slight downstream movement again. However, we define the quenching distance d_q as the smallest distance on the quenching curve, indicated with a red dot in Fig. 6.

Fig. 7 compares the minimum $\dot{\omega}_{H_2}$ along the symmetry axis for different starting locations of the steady-state flame. To modify the initial location of the flame, different inlet velocities U_{ini} were used for steady-state simulations. The initial velocities that were investigated are, respectively, 7 m/s, 9 m/s and 11 m/s. All cases show a similar trend. The initial velocity and the corresponding initial location of the flame affect only the initial phase of the flame path. For high U_{ini} values, the flame stabilizes in a location with a larger channel width and the flame has a higher hydrogen consumption rate. After the initial phase, all three flames with different U_{ini} quenches at nearly the same quenching distance (0.7 percent deviation between 7 m/s and 11 m/s), showing that the starting location of the flame does not affect the quenching distance for this study.

The effect of final velocity U_f, which determines the traveling velocity of the flame during the transient phase, is also investigated for the

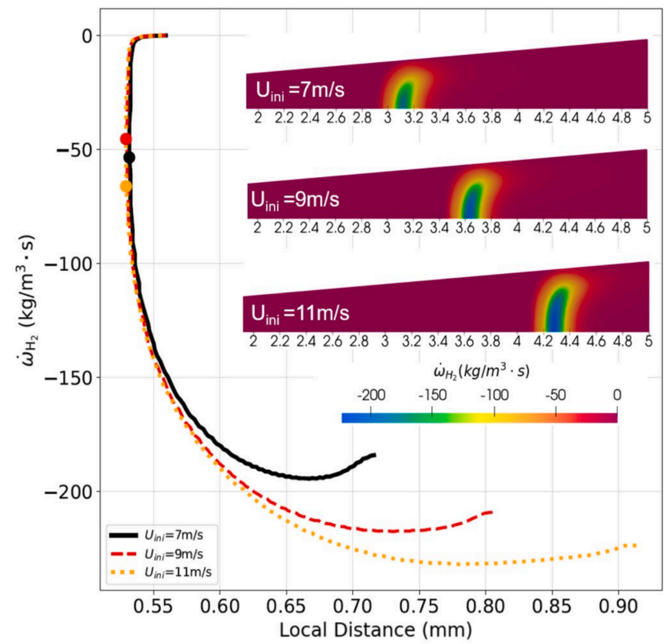

Fig. 7. Effect of Uini on quenching distance for H_2-air flame, $\phi = 1$ at T = 300 K.

CH_4 and H_2 flames. A final velocity $U_f = 0$ means that there is no mixture flowing into the domain, which represents a complete shut-off of the fuel–air supply. Fig. 8 compares the quenching curves of 3 different U_f values for stoichiometric H_2-air flames with $T_{\text{wall}} = 300$ K using the slot configuration (A). Although the trend is the same for all mixtures, the flame quenching distance is significantly affected by U_f.

The quenching distances (d_q) for different equivalence ratios are presented and discussed in the next section. The same procedure presented in this section is followed for CH_4-air mixtures with equivalence ratio varying from 0.6 to 1.3 and for H_2-air mixtures with ϕ from 0.5 to 2.4.

4. Results and discussion

This section is divided into two main parts, where CH_4 and H_2 flame

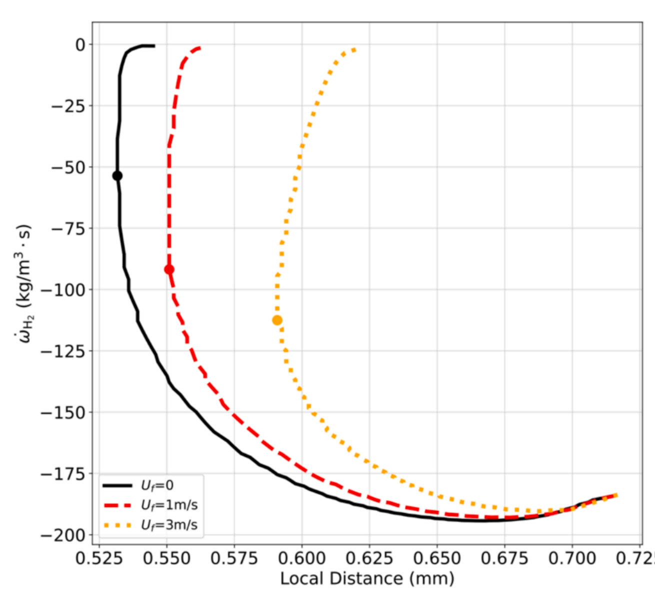

Fig. 8. Effect of Uf on quenching distance for H_2-air flame, $\phi = 1$ at T = 300 K.

results will be presented separately. Asexplained in the previous sections, the modeling setup and methodology are the same for both fuels.

4.1. Methane-air

In this section, numerically determined quenching distances will be presented for CH_4-air flames. The effect of geometry and U_f is discussed, and the results are compared with the experiments found in the literature.

4.1.1. Effect of final velocity on quenching distance

In this section, d_q results are presented for the slit configuration with the equivalence ratio ϕ ranging from 0.7 to 1.3 in increments of $\Delta\phi = 0.1$, and compared with the experimental values found in the literature. Fig. 9 presents the d_q results, where the x-axis represents ϕ and the y-axis represents the d_q value in mm. A fourth-order polynomial is fitted to the obtained d_q values and represented by lines for both values of U_f. The polynomial fits for different values of U_f are provided in Table 2. The slit configuration is used for comparison with experiments, as the experimental methods used as references can be best represented by the slit configuration, except for the ASDT method [19,20,21], which can best be represented by the annular duct configuration. However, in the next section, we show that the slit and annular duct configurations result in approximately the same d_q values. Only the parallel plate configuration in the work of Harris et al. [17] is reported in Fig. 9 for comparison. The value of $U_f = 0.25$ m/s was selected to allow direct comparison with the experimental results of Liu et al. [19,20] as this velocity corresponds to the flow rate of 2 L/min used in their experiments.

Both the numerical d_q results and the experimental values found in the literature follow the same trend for varying ϕ values. The minimum quenching distance is observed at stoichiometric conditions for both U_f.

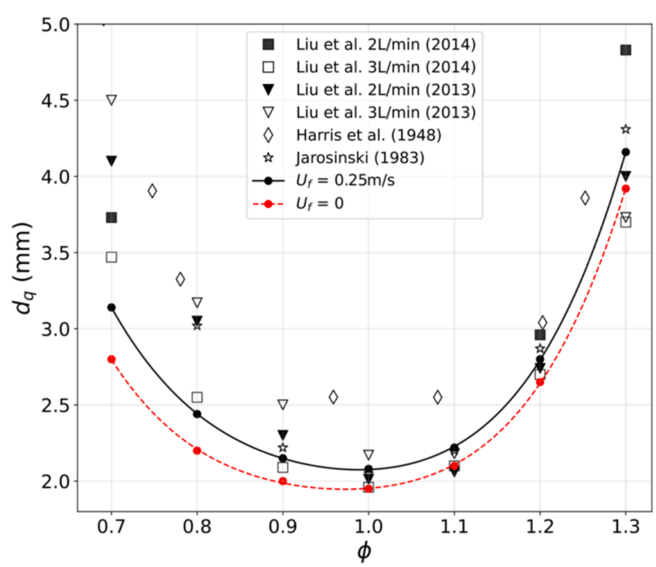

Fig. 9. CH_4-air quenching distances as function of equivalence ratio ϕ for different final velocity Uf.

The d_q increases toward both the lean and rich sides. The numerical results show good agreement with the d_q values found in the literature, especially for stoichiometric and rich conditions. The discrepancy between the obtained d_q values in this work and the experimental results increases toward the lean side. This difference could be associated with the chemical mechanism's ability to predict quenching, but it should also be noted that the discrepancy between the different experimental datasets increases toward the lean side. In the ASDT setup [19,20], the annular tube is composed of discrete steps, which means that the passage is not completely smooth. Moreover, in the work of Jarosinski [18], it was shown that the flame front was disturbed in the experiments, and the flame front lost its perfectly concave shape. Since lean flames are more prone to be affected by such perturbations, this may influence the final quenching distance results. Furthermore, the difference might be due to the definition of quenching distance used in this work. In this work, the net production rate ($\dot{\omega}_i$) is used to locate the flame front. In the experiments, the flame front is detected with the Schlieren method [18] or directly with the CCD camera [19,20]. Thus, these differences in locating the flame front can contribute to the deviation between the d_q results.

From Fig. 9, it is clear that d_q decreases with decreasing U_f. In the experimental work of Liu et al. [19,20], it is also observed that the final flow rate affects the quenching distance. However, in [19,20], it was stated that the deviation caused by the final flow rate is negligible. However, in this work, a clear relation between the quenching distance and the final velocity is presented. The decrease in quenching distances with decreasing inlet velocity can be attributed to the flame propagation speed (v_f). On the symmetry axis, the flame propagation speed can be calculated roughly as: $v_f = Sl - U_{axis}$ where U_{axis} is the velocity on the symmetry axis, determined by U_f and the local wall-to-wall distance. For higher U_f, the flame travels at a lower speed towards the inlet since the laminar flame speed is nearly independent of U_f. Thus, there is more time for heat transfer from the flame to the wall in the critical region. Therefore, the flame will quench earlier at a location with a wall-to-wall distance that is slightly larger. It is important to note that d_q measurements depend on U_f, which should be taken into account when designing an experiment or product.

4.1.2. Effect of geometry on quenching distance

Three different configurations were investigated numerically: a rectangular slit (A), a circular duct (B), and an annular duct (C), and the resulting quenching distances are compared. For all three configurations, the same value for the inlet velocity $U_f = 0$ was used. Fig. 10 presents the quenching distances obtained as a function of the equivalence ratio. A fourth-order polynomial is again fitted to the obtained d_q values. The polynomial fits for the CH_4-air flames in different geometries are provided in Table 2.

From Fig. 10, it can be observed that the quenching distance results for the slit and annular duct configurations are nearly identical, with the annular duct predicting a quenching distance that is only around 3 percent larger compared to the planar slit. This is due to the large inner diameter of the annular passage compared to the clearance between the tubes, and therefore we conclude that the experiments of Liu et al. [19,20] accurately represent a nearly 2D planar geometry.

In the case of the circular duct, the quenching distance is 16–40 %

Table 2
Polynomial fits of dq in mm as function of ϕ for CH_4-air for different geometries and final velocity Uf.

Geometry	U_f	Polynomial Fit
Slit	0	$d_q(\phi) = 78.41\phi^4 - 297.25\phi^3 + 431.72\phi^2 - 285.11\,\phi + 74.31$
Slit	0.25 m/s	$d_q(\phi) = 76.14\phi^4 - 288.71\phi^3 + 418.11\phi^2 - 274.16\,\phi + 70.06$
Annular Duct	0	$d_q(\phi) = 118.48\phi^4 - 443.27\phi^3 + 632.02\phi^2 + 407.34\phi + 102.74$
Circular Duct	0	$d_q(\phi) = 81.41\phi^4 - 309.92\phi^3 + 451.41\phi^2 - 298.53\phi + 77.75$

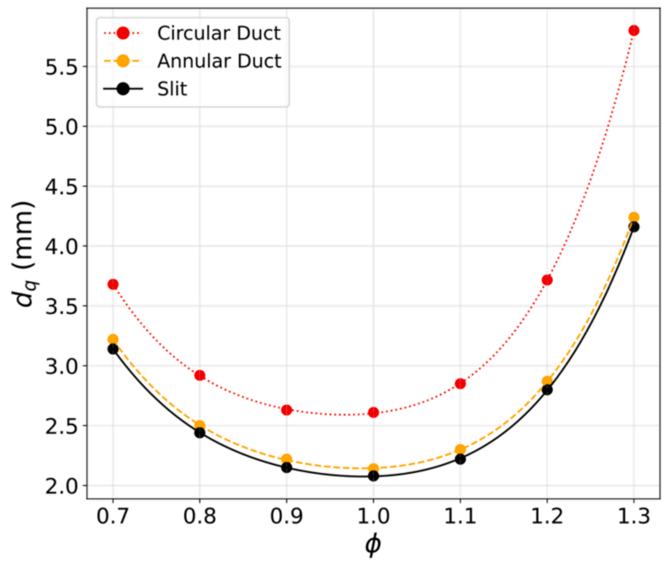

Fig. 10. Fig. 10. CH_4-air quenching distances as function of equivalence ratio ϕ for different geometries.

higher compared to the planar configuration where the difference increases towards the rich side. This behaviour is also observed in the work of Harris et al. [17] for methane air and propane air mixtures where parallel plates and cylindrical tubes were used to determine the quenching distances. This difference may be partly due to the increase in the contact surface area between the flame and the walls.

In the following Eqs. (8)–(10), the ratio (σ) of the cross-sectional area (A) to the wetted perimeter (P), is expressed for the different configurations:

$$\sigma_{\text{slit}} = \frac{A_{\text{slit}}}{P_{\text{slit}}} = \frac{(h \times w)}{2h + 2w} \approx \frac{w}{2} = \frac{d}{2} \quad \textit{assuming } w \ll h \tag{8}$$

$$\sigma_{\text{circular}} = \frac{A_{\text{circular}}}{P_{\text{circular}}} = \frac{\left(\frac{\pi D^2}{4}\right)}{\pi D} = \frac{D}{4} = \frac{d}{4} \tag{9}$$

$$\sigma_{\text{annular}} = \frac{A_{\text{annular}}}{P_{\text{annular}}} = \frac{\left(\frac{\pi\left(D_o^2 - D_i^2\right)}{4}\right)}{\pi(D_o + D_i)} = \frac{\left(\frac{\pi\left((D_i + 2d)^2 - D_i^2\right)}{4}\right)}{\pi((D_i + 2d) + D_i)} = \frac{4D_i d}{4(2D_i + 2d)} \approx \frac{d}{2} \tag{10}$$

For all configurations, the distance between the upper and lower wall (d) is considered equal, which corresponds to the slit width (w) for the slit configuration, the diameter (D) of the circular duct, and the distance between the outer and inner diameter of the annular duct ($\frac{D_o - D_i}{2}$). The comparison is made based on the assumption that the slit height (h) is much greater than the slit width (w). It is evident that this ratio is twice as small for a circular duct compared to a slit, and it is nearly identical for the slit and annular duct. This suggests that for the same diameter, the circular duct configuration offers twice the surface area for heat transfer. This difference in contact surface area has a first-order effect on d_q. However, since the d_q difference between the circular duct and the rest is not constant with changing equivalence ratio, the change in contact surface area cannot be the only reason. Secondary effects such as flame curvature near the wall, flame stretch, and velocity profile also effect d_q.

4.2. Hydrogen-air

In this section, the quenching of H_2-air flames will be discussed.

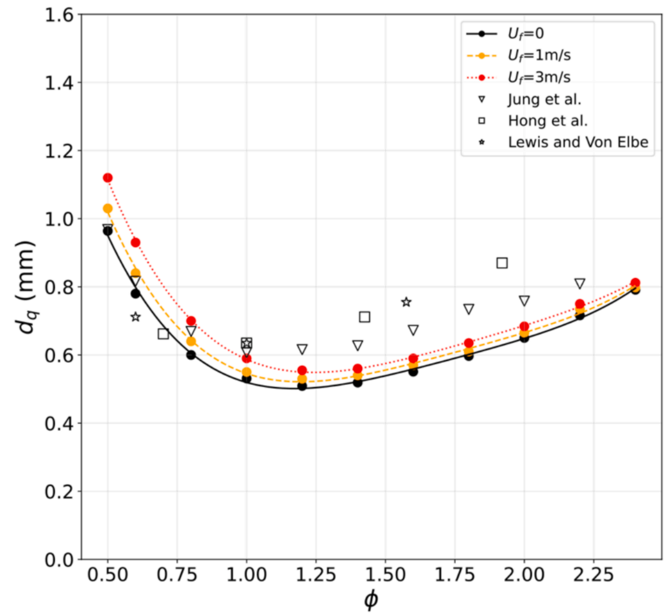

Fig. 11. H_2-air quenching distances as function of equivalence ratio ϕ for different final velocity Uf.

Since the quenching distances are smaller for H_2-air than CH_4-air mixtures, the computational domains were chosen to be smaller to reduce computational costs. The effects of U_f, geometry, and wall temperature will be presented next.

4.2.1. Effect of final velocity on quenching distance

Quenching distance results of H_2-air mixtures were obtained for equivalence ratios ranging from 0.5 to 2.4, and presented in this section. Fig. 11 shows the d_q values versus equivalence ratio for three different U_f values: 0, 1, and 3 m/s, respectively. The coefficients of the polynomial fits are provided in Table 3. The d_q values are compared with experimental results found in the literature [5,16,21], which were obtained using different techniques. The results from this study generally show good agreement with the experimental data in the literature. The quenching distance is highest on the lean side, with the minimum d_q observed at $\phi = 1.2$ with a value of 0.52 mm. As the mixture becomes richer, d_q starts increasing again. This trend is consistent with both the experimental results found in the literature. As the final velocity increases from 0 to 3 m/s, the quenching distance values also increase. This behavior is similar to that observed in CH_4-air flames; however, the difference between the quenching results of H_2-air flames at various velocities is not as pronounced as in CH_4-air flames. As shown in Fig. 11, the deviation increases towards the lean side of the curve.

With increasing U_f, the quenching distance increases almost linearly, which is illustrated in Fig. 12. It has seen that, the sensitivity is larger for the lean flames, and the effect of final velocity on d_q becomes less severe for rich mixtures. One of the possible explanation for this phenomenon is the differences in flame propagation speed (v_f) upstream, as discussed in detail in 4.1.1. For mixtures with lower Sl values, the flame propagation speed will be lower for the same U_f, which results in more residence time

Table 3
Polynomial fits of d_q in mm as function of ϕ for H_2-air for different geometries and final velocities U_f.

Geometry	U_f	Polynomial Fit
Slit	0	$d_q(\phi) = 2.48 \times 10^{-1}\phi^4 - 1.72\phi^3 + 4.51\phi^2 - 5.08\phi + 2.56$
Slit	1 m/s	$d_q(\phi) = 2.47 \times 10^{-1}\phi^4 - 1.75\phi^3 + 4.66\phi^2 - 5.35\phi + 2.73$
Slit	3 m/s	$d_q(\phi) = 2.24 \times 10^{-1}\phi^4 - 1.64\phi^3 + 4.55\phi^2 - 5.43\phi + 2.88$
Annular Duct	0	$d_q(\phi) = 2.33 \times 10^{-1}\phi^4 - 1.65\phi^3 + 4.42\phi^2 - 5.05\phi + 2.58$
Circular Duct	0	$d_q(\phi) = 1.65 \times 10^{-1}\phi^4 - 1.19\phi^3 + 3.33\phi^2 - 3.82\phi + 2.28$

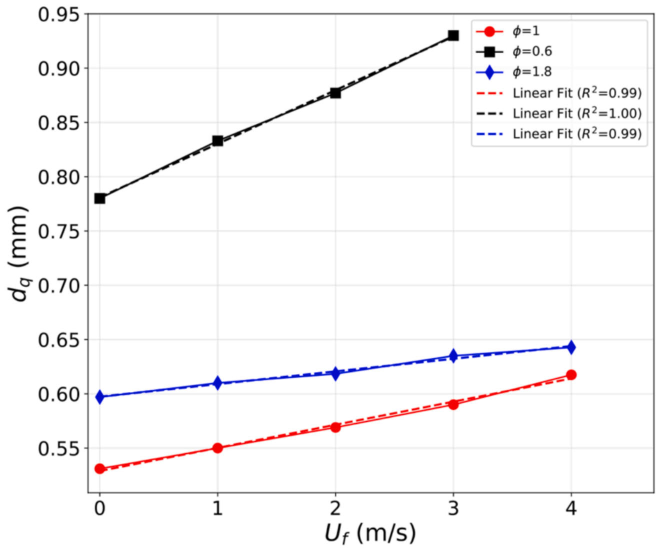

Fig. 12. Effect of Uf on quenching distance for H_2-air flame with various ϕ at T = 300 K. Dashed lines represent linear fits.

Table 4
Effect of the slope angle α (in degrees) on the quenching distance of H_2-air flames, ϕ = 1 and T_{wall} = 300 K.

U_f (m/s)	Quenching distance (mm)		
	$\alpha = 2.4$	$\alpha = 4.8$	$\alpha = 9.6$
0	0.54	0.53	0.53
1	0.56	0.55	0.55
2	0.57	0.57	0.57

in the critical region. Up to $\phi = 1.8$, the laminar burning velocity Sl of H_2-air flames increases with increasing equivalence ratio. Thus, the flames with lower Sl, in this case lower equivalence ratio, will experience a larger influence of U_f.

For H_2-air flames, all d_q results reported in this manuscript were performed with a 4.8 degree slope angle. The effect of the slits slope angle on d_q is also investigated by halving and doubling the angle, and the results are presented in Table 4 for $\phi = 1$. When the slope angle increased from 4.8 degrees to 9.6 degrees, no effect was observed on d_q for all three values of U_f. A slight increase in d_q was observed when the slope angle decreased to 2.4 deg from 4.8 deg for $U_f = 0$ and 1 m/s. This implies that for slope angles that are small enough, the angle does not affect d_q significantly. If the slope angle changes drastically to a value such as 45 degrees, it could be expected to see a greater influence on d_q due to a sudden change in the velocity field. Since it is difficult to accurately determine d_q with a high slope angle, due to the rapid change in diameter with a small displacement in the x direction, the authors did not investigate slope angles greater than 9.6 degrees.

4.2.2. Effect of geometry on quenching distance

Quenching distances are obtained for different configurations and are compared in Fig. 13. All computations are performed with $U_f = 0$. The polynomial fits for H_2-air flames with different geometries are given in Table 3. As observed in CH_4-air mixtures, the slit and annular duct configurations give similar results, which is consistent with the 2D slit assumption of ASDT experiments performed by Jung et al. [21]. Higher d_q values are observed in the circular duct configuration compared to others for each equivalence ratio. It is also clearly seen that the deviation between the slit and circular duct configurations increases towards the rich side. While for $\phi = 0.5$ it is only a 14 percent increase in quenching distance, this ratio increases to 62 percent for $\phi = 2$, where it reaches the maximum value within the range that was investigated. As discussed for

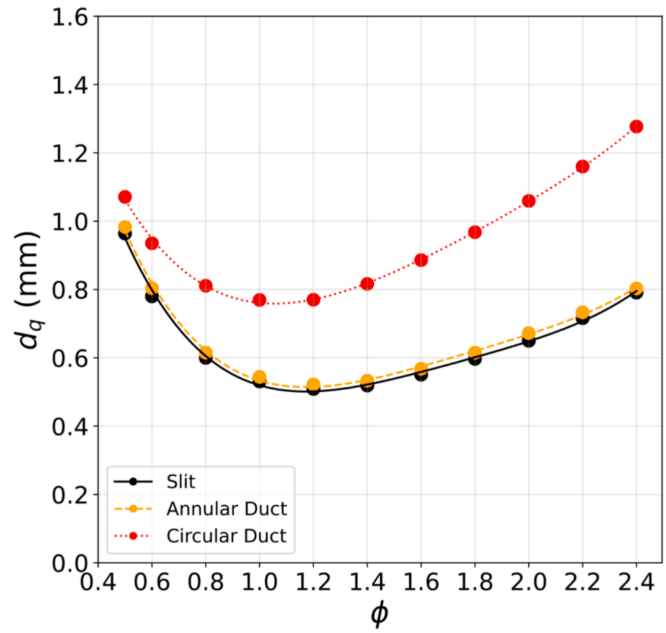

Fig. 13. H_2-air quenching distances as function of equivalence ratio ϕ for different geometries.

CH_4-air mixtures, this difference cannot be related only to contact surface area. Flame stretch, curvature and effect of differential diffusion might be relevant for this phenomenon, which should be investigated in the future.

4.2.3. Effect of wall temperature on quenching distance

The effect of wall temperature on the quenching distance of H_2 air mixtures is investigated and presented in this section. The isothermal wall temperature (T_{wall}) and inlet temperature of the premixture are increased from 300 K to 800 K with 100 K increments, but only for the slit configuration. The inlet temperature is chosen to be the same as the wall temperature to model the inlet mixture as a fully developed flow in thermal equilibrium. Fig. 15 presents d_q vs. wall temperature results for several ϕ values. As the wall temperature increases, quenching distance decreases. As discussed in detail in the 3, the main reason behind the quenching is heat loss to the cold walls. This heat loss gets larger when the temperature difference between the flame and wall increases. Thus, the wall temperature is expected to have a significant effect on the quenching distance.

Fig. 14 a) shows the comparison of the temperature contours for $T_{wall} = T_{inlet} = 300$ K (top) and 500 K (bottom) in the quenching instance for $\phi = 1$. It is seen that when the wall and inlet temperature are increased, the flame can penetrate further and quenches at a smaller wall-to-wall distance. The difference between the flame temperatures in the instance of quenching is negligible (7 K), even though the wall temperatures differ by 200 K. This implies that the quenching distance is highly influenced by the flame temperature needed for the activation of the chemical reactions. Fig. 14 b) shows the comparison of $\dot{\omega}_{H_2}$ for the same conditions. One can notice that for the 500 K case, $\dot{\omega}_{H_2}$ is not negligible near the wall, indicating that some reactions are still taking place and H_2 is consumed near the wall due to the relatively high wall temperature.

The relation between d_q and T_{wall} is found to be nearly linear for the investigated conditions, which is illustrated in Fig. 15. A global equation that can be used to obtain the quenching distance of H_2-air mixtures in the range of $\phi = 0.5$ to 2.4 and $T_{wall} = 300$ K to 800 K, is given by

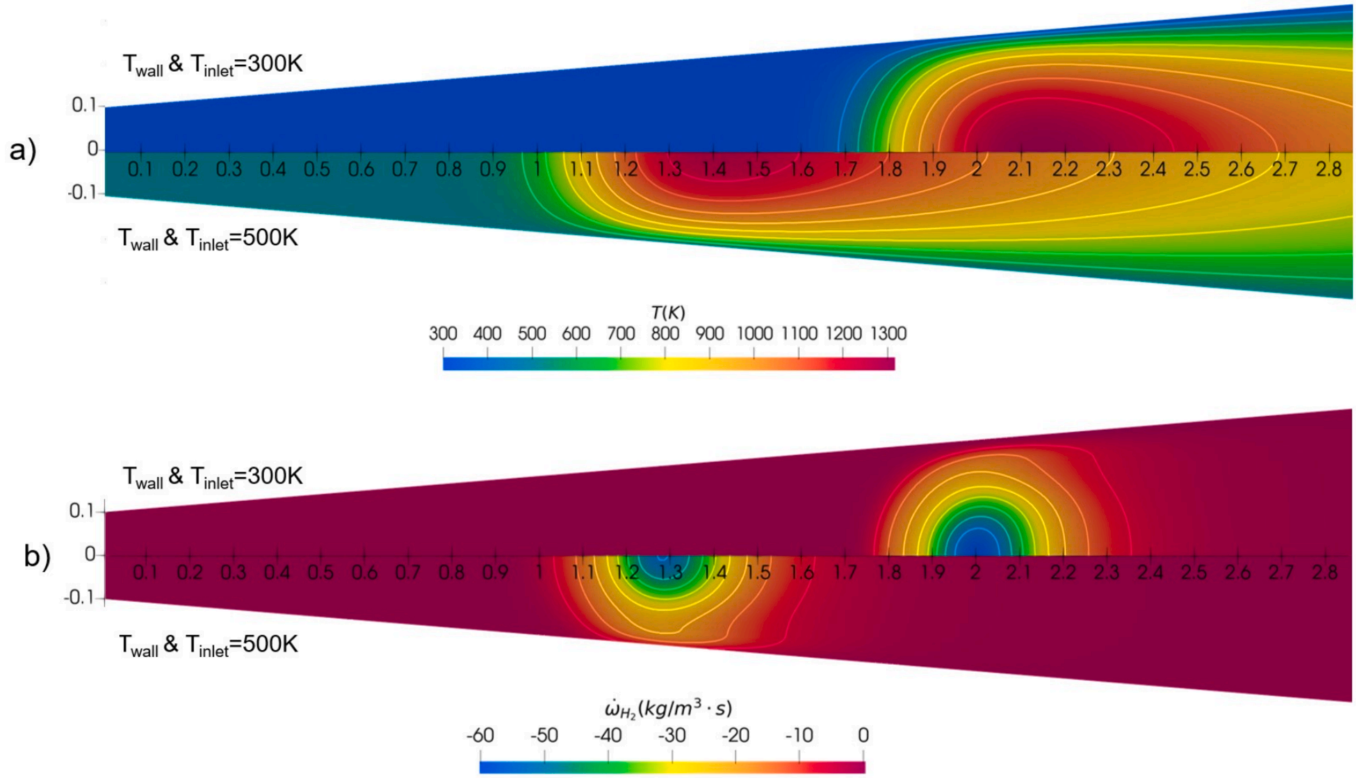

Fig. 14. A) temperature (k) and b)$\dot{\omega}_{H_2}$(kg/m^{3}s) contours at the quenching instance of H_2-air flames with $\phi = 1$. Results for $T_{wall} = 300$ K and 500 K are shown in the top and bottom half of each plot, respectively.

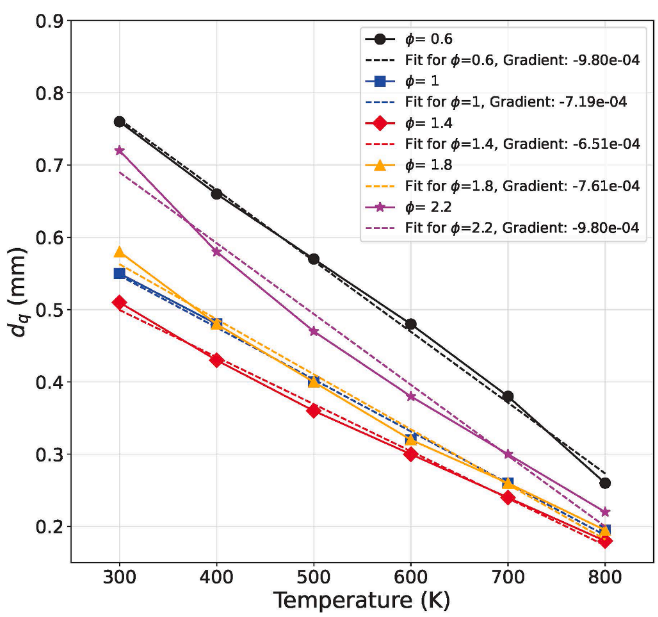

Fig. 15. H_2-air quenching distances as function of temperature for different equivalence ratio ϕ. Dashed lines represents linear fits.

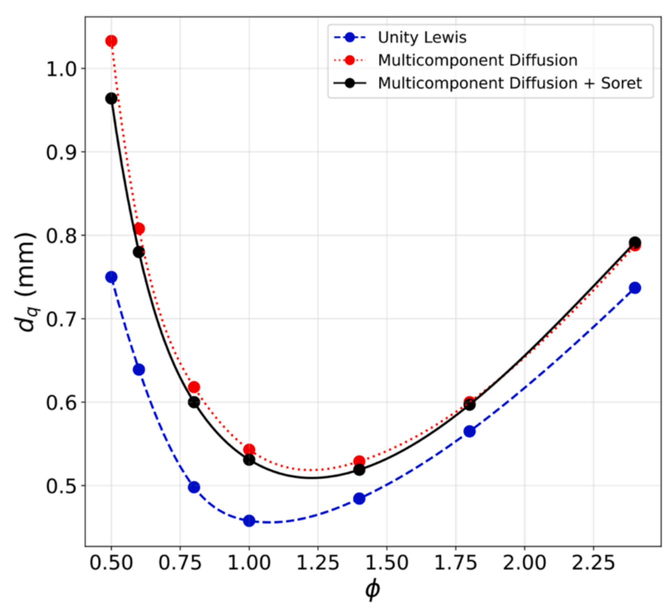

Fig. 16. H_2-air quenching distances as function of equivalence ratio ϕ for different diffusion models.

$$d_q(\phi, T) = (-0.0045549\phi + 0.0069658\phi^2 - 0.0038594\phi^3 + 0.00070112\phi^4)T + 0.90763\phi - 2.7266\phi^2 + 1.8015\phi^3 - 0.34764\phi^4 + 1.1436 \tag{11}$$

This equation is a fit to the data in Fig. 15 with $R^2 = 0.994$.

For the conditions studied in this work, 800 K is found as the limiting temperature for quenching distance calculations. Above this temperature, it is hard to detect the flame front and the quenching distance for H_2-air mixtures. When simulations are performed above 800 K, the reaction zone becomes too wide and the flame front becomes ill defined. This can be associated with autoignition of H_2-air mixtures, which is reported to occur at 853 K in the literature [39]. When we tried to model quenching at 900 K, we found that the flame will not quench even when the gap is an order of magnitude smaller than for the 800 K case.

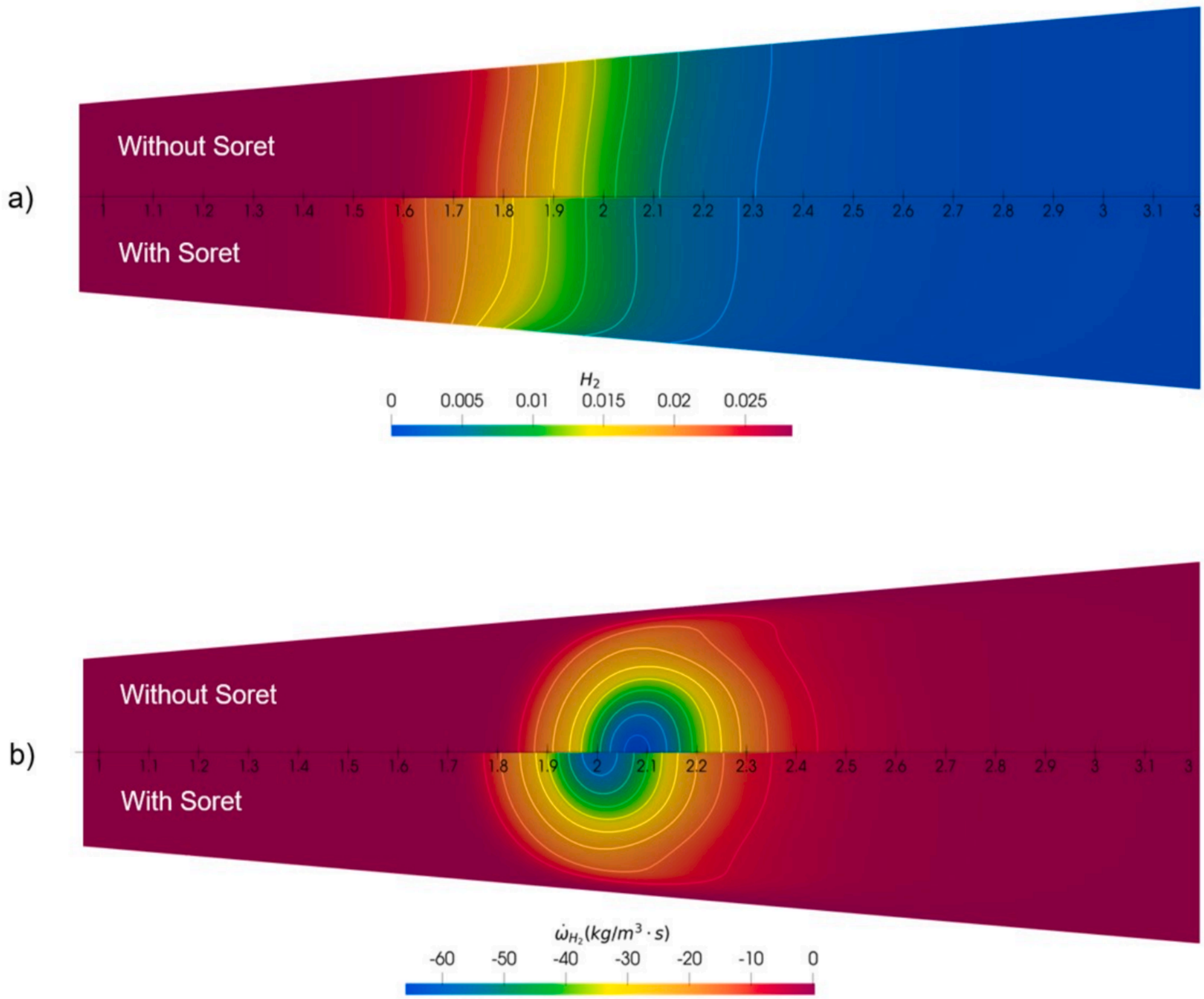

Fig. 17. Comparison of diffusion modeling; without Soret effect (top half) and with Soret effect (bottom half) at the quenching instance ($\phi = 1$, $T_{wall} = 300$ K), a) H_2 mass fraction and b) $\dot{\omega}_{H_2}$ (kg/m^3s).

4.2.4. *Effect of multicomponent diffusion modeling*

The effect of diffusion modeling is investigated for H_2-air flames using the slit configuration and results are presented in Fig. 16. Quenching distances presented in the previous sections are compared to computations with the multicomponent diffusion approach but without Soret effect, and to simulations employing the unity Lewis number approach without Soret effect.

Fig. 17 a) shows the comparison of H_2 mass fraction contours computed without Soret effect (top) and with Soret effect (bottom) at the quenching instance ($\phi = 1$). It can be seen that H_2 diffuses away from the wall towards the flame due to the temperature gradient when the Soret effect is included. This results in a gradient of H_2 mass fraction normal to the wall, which is absent when thermodiffusion is not included. Furthermore, it can be observed that the H_2 mass fraction is more gradually decreasing due to thermodiffusion along the symmetry axis, especially on the unburned side. Fig. 17 b) shows the comparison of $\dot{\omega}_{H_2}$ contours for the same cases. When thermodiffusion is included, the flame quenches at a slightly smaller distance. This might be associated with a slightly stronger flame core near the symmetry axis due to a higher H_2 diffusion flux. Anyway, it can be concluded that there is only a slight difference in quenching distance when the Soret effect is included.

Fig. 16 shows that when the Soret effect is included, d_q slightly decreases for mixtures with ϕ smaller than 1.8. The effect of the Soret effect on d_q was also investigated for $T_{wall} = 500$ K and a similar relationship was found to that of the case $T_{wall} = 300$ K. In a previous numerical one-dimensional head-on quenching study [40], it has been shown that the quenching distance of H_2-NH_3 blends increases when the Soret term is included. For a head-on quenching configuration, this is expected, as the flame weakens near the wall and quenches at an earlier stage. This also means that the effect of thermodiffusion on quenching depends on the

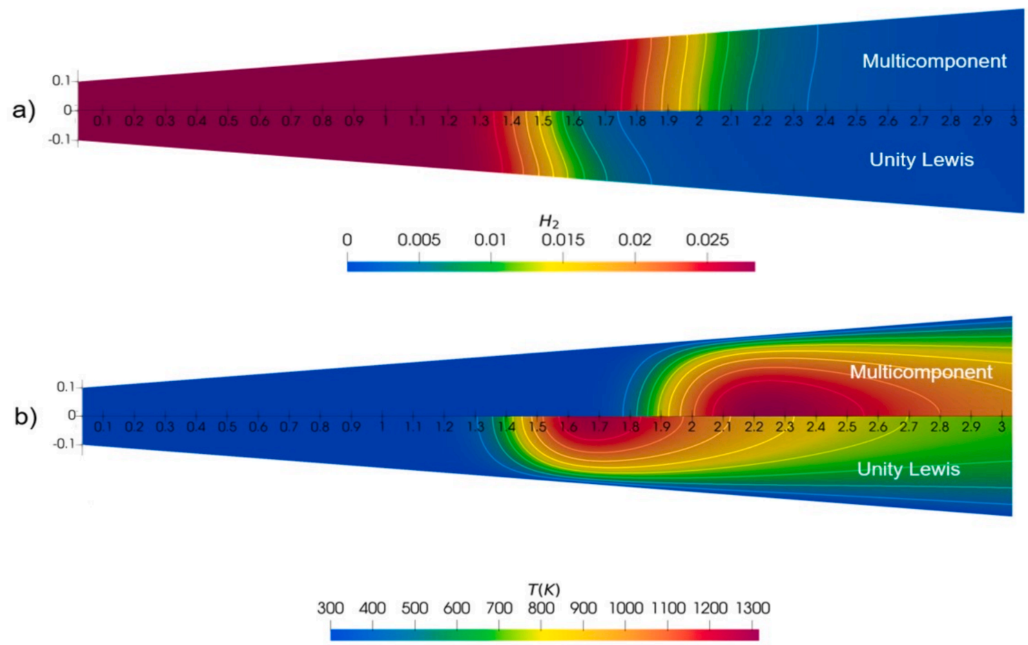

Fig. 18. The comparison of diffusion modeling with multicomponent diffusion (top half) and unity Lewis number assumption (bottom half) at quenching instance ($\phi = 1$, $T_{wall} = 300$ K) a) H_2 mass fraction contour, b) Temperature (K) contour.

configuration.

The unity *Le* approach significantly affects d_q. Compared to the multicomponent diffusion results, d_q decreases when the unity *Le* approach is used, and the difference increases with decreasing ϕ since *Le* is moving away from unity. The Lewis number of H_2 is much lower than 1, which means its mass diffusivity is higher than the thermal diffusivity of the mixture. When the quenching distances found by the two methods were divided by the flame thicknesses computed with the same methods, the agreement between the two methods does not improve much. In fact, dividing by the flame thickness makes the discrepancy worse on the rich side. Therefore, the difference in quenching distances cannot be explained only by the difference in flame thickness. Fig. 18 a) shows the comparison of H_2 results computed with multicomponent diffusion (top) and with unity Lewis number assumption (bottom) at the quenching instance ($\phi = 1$). The Soret effect is not included for both cases. When diffusion is modeled with the unity Lewis number assumption, quenching occurs at a smaller distance for the same mixture.

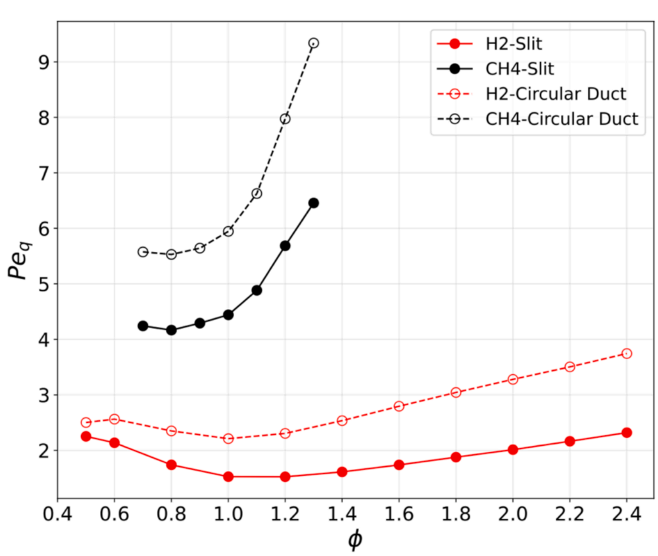

Fig. 19. Quenching Peclet numbers Pe_q as function of equivalence ratio ϕ for different fuels and geometries.

This is expected due to the high diffusivity of H_2. Fig. 18 b) shows the comparison of temperature contours for the same conditions. When the flame temperatures are compared at the quenching instance, the difference between the flame temperatures is negligible, as was observed in the comparison of cases with different wall temperatures. This phenomenon reinforces the idea that quenching is highly associated with the temperature of the flame. When the flame reaches below a certain temperature, for this case 1300 K, the quenching occurs independently of the diffusion model or the temperature of the wall. From the figure, one can see that when multicomponent diffusion effects are included, H_2 spreads more along the symmetry axis compared to the case with the unity Lewis number assumption.

4.3. Comparison of methane and hydrogen quenching distances

One of the important outcomes of this study is to point out the differences in the quenching distances and the quenching behavior of CH_4-air and H_2-air flames for the design of fully hydrogen-fueled burners. To compare quenching distances of different fuels, the non-dimensional quenching Peclet number is used widely in the literature [41,42] which is defined as:

$$Pe_q = d_q/\delta_F. \tag{12}$$

In this study, the thermal flame thickness definition [14] is used and is based on the simulation of a 1D freely propagating flame.

This finding contradicts [18], which concludes that the ratio of quenching distance to flame thickness obtained from the experimental work of Bradley et al. [43] is almost constant with a value of 2.0 to 2.2. The quenching distances obtained in this study, as illustrated in Fig. 19, align closely with the results reported by [18]. Therefore, the difference in the ratios can be attributed to differences in the flame thickness. For lean and stoichiometric CH_4-air flames, Pe_q is around 4.2 for the slit configuration and 5.6 for the circular duct. On the rich side, Pe_q increases, indicating that the quenching distance increases faster than the thickness of the flame. For H_2-air flames, Pe_q is significantly lower than for CH_4-air flames. The lowest Pe_q is observed around stoichiometry for both configurations. As the mixture gets richer, Pe_q increases almost linearly. As observed for the CH_4-air flames, Pe_q is not constant over the investigated ϕ range. It should be noted that the quenching Peclet number is not defined in the same way in different studies, mainly due to the difference in the definition of flame thickness [18,21,44,45]. Thus, the aim in this study is not to compare the values obtained with the literature, but to compare H_2 and CH_4-air mixtures in a non-dimensional way.

5. Conclusions

This numerical study has provided an in-depth analysis on quenching distance determination of laminar premixed H_2-air and CH_4-air flames. The quenching behavior is modeled using finite-rate chemistry with transient simulations. Quenching distances are obtained for a wide range of equivalence ratios for both mixtures, and engineering correlations are presented in that range. The research revealed that the quenching distance is notably affected by the inlet speed of the fuel–air mixture for the slit configuration. Specifically, a decrease in inlet speed results in a larger flame traveling velocity and a smaller quenching distance. This is an important effect that should be taken into account in flash-back and quenching analyses. The effect of different geometries on the quenching distance was also explored, including slits, circular ducts, and annular ducts. The quenching distance for the slits and annular ducts are nearly the same, which validates the 2D assumption of previous experiments using this setup. However, in the case of the circular duct, the quenching distance is found to be higher than that of the other configurations for both fuels. Thus, this study concludes that quenching distance is highly dependent on the setup and should be addressed while designing an experimental setup. Furthermore, the effect of wall temperature is investigated for H_2-air flames and a linear relationship between the wall temperature and the quenching distance was found. As the wall temperature increases, the quenching distance decreases due to decreased heat transfer to the wall. The importance of diffusion modeling is highlighted for H_2-air premixtures and it is concluded that Soret diffusion does not have a significant effect on the quenching distance results, while the unity Lewis number assumption predicts the quenching distances poorly. Finally, non-dimensionalized quenching distances are compared and H_2 flames are found to be able to penetrate smaller gaps without quenching compared to their flame thickness than CH_4 flames.

CRediT authorship contribution statement

Tahsin Berk Kiymaz: Writing – original draft, Methodology, Data curation, Conceptualization. **Nijso Beishuizen:** Writing – review & editing, Supervision, Conceptualization. **Jeroen van Oijen:** Writing – review & editing, Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. . Comparison of different chemical mechanisms

Figs. A20 and A21 show quenching distances as a function ϕ for methane and hydrogen flames, respectively, calculated with different reaction mechanisms.

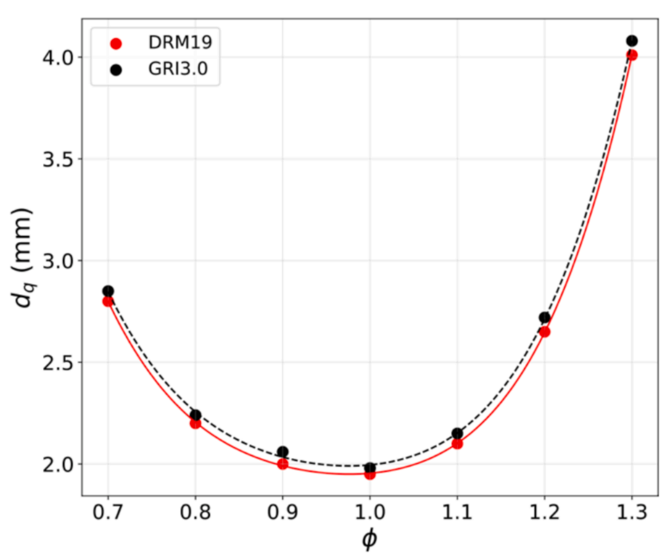

Fig. A20. CH_4-air quenching diameters for slit configuration, comparison of GRI 3.0 [25] and DRM 19 [26] mechanisms.

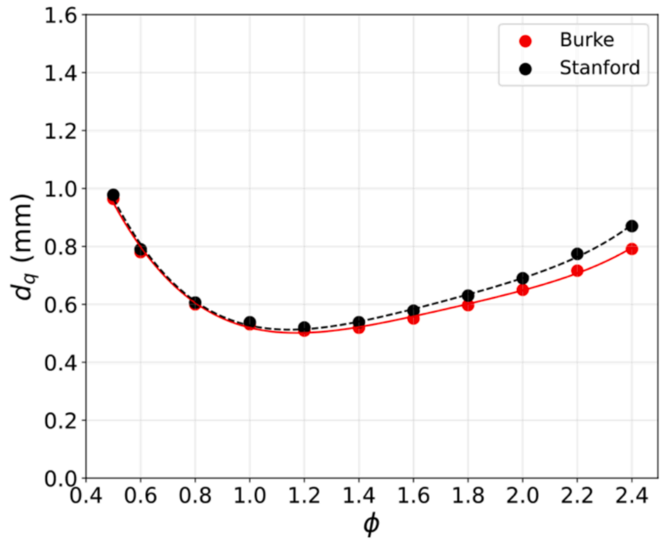

Fig. A21. H_2-air quenching diameters for slit configuration, comparison of Burke [27] and Stanford mechanisms [28].

Data availability

Data will be made available on request.

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